

Advanced (Habanero)

Q1. Factor the expression: $x^2 + 5x + 4x + 20$

- (A) $(x + 4)(x + 5)$
- (B) $(x + 2)(x + 10)$
- (C) $(x + 1)(x + 20)$
- (D) $(x + 3)(x + 15)$
- (E) $(x + 6)(x + 10)$

Q2. Factor the expression: $3x^2 + 12x + 4x + 16$

- (A) $3(x + 2)(x + 4)$
- (B) $3(x + 4)(x + 2)$
- (C) $4(x + 2)(x + 3)$
- (D) $3(x + 3)(x + 4)$
- (E) $2(x + 3)(x + 6)$

Q3. Factor the expression: $x^2 + 2x + 8x + 16$

- (A) $(x + 2)(x + 8)$
- (B) $(x + 8)(x + 2)$
- (C) $(x + 4)(x + 4)$
- (D) $(x + 1)(x + 16)$
- (E) $(x + 16)(x + 1)$

Q4. Factor the expression: $2x^2 + 5x + 3x + 15$

- (A) $(2x + 3)(x + 5)$
- (B) $(2x + 5)(x + 3)$
- (C) $(2x + 15)(x + 1)$
- (D) $(x + 3)(2x + 5)$
- (E) $(x + 15)(2x + 1)$

Q5. Factor the expression: $3x^2 + 12x + 2x + 8$

- (A) $3(x + 2)(x + 4)$
- (B) $3(x + 4)(x + 2)$
- (C) $(3x + 2)(x + 4)$
- (D) $(3x + 4)(x + 2)$
- (E) $2(3x + 2)(x + 2)$

Q6. Factor the expression: $2x^2 + 7x + 5x + 35$

- (A) $(2x + 5)(x + 7)$
- (B) $(2x + 7)(x + 5)$
- (C) $(x + 7)(2x + 5)$
- (D) $(x + 5)(2x + 7)$
- (E) $(2x + 35)(x + 1)$

Q7. Factor the expression: $x^2 + 6x + 5x + 30$

- (A) $(x + 5)(x + 6)$
- (B) $(x + 6)(x + 5)$
- (C) $(x + 1)(x + 30)$
- (D) $(x + 30)(x + 1)$
- (E) $(x + 6)(x + 10)$

Q8. Factor the expression: $4x^2 + 20x + 5x + 25$

- (A) $5(4x + 5)(x + 1)$
- (B) $5(x + 1)(4x + 5)$
- (C) $5(x + 5)(4x + 1)$
- (D) $5(4x + 1)(x + 5)$
- (E) $4(x + 5)(x + 5)$

Open Ended Questions (FRQ)

Q9. Factor the expression: $3x^2 + 10x + 8x + 40$. Verify your answer by multiplication.

Q10. Factor the expression: $2x^2 + 7x + 3x + 21$. Verify your answer by multiplication.

Chef's Tips

- Always verify factoring by multiplication
- Use the distributive property for multiplication
- Pay attention to the signs of terms when factoring